Fishing for Sustainability — Final Debrief & Synthesis

From Carrying Capacity to MSY and Management Tradeoffs

Name: ______________________________
Date: ____________________

Quick Recap (fill in from your previous sheets)

Round	Strategy	Total Catch	Fish Remaining After Harvest	Fish After Reproduction
1	Open Access			
2	Fixed Quota			
3	Fixed Effort			

Key Biological Concepts
- **Carrying Capacity (K):** Maximum population size the environment can sustain.
- **Reproduction Rule Used:** After harvest, the population **doubles** up to a cap of **30** (our K).
- **Maximum Sustainable Yield (MSY):** The largest long-term average catch that can be taken without depleting the population (often near ~½ K in simple logistic models).

Individually Answer (short responses)

1. **MSY & K:** Given our reproduction rule and cap at 30, explain why the sustainable catch per round is expected to be highest when the stock left after harvest is near the midpoint of capacity.

2. **Compare Strategies:** In one or two sentences each, give **one advantage** and **one drawback** for:
   - Open Access
   - Fixed Quota
   - Fixed Effort

3. **Fairness vs. Sustainability:** Which strategy best balanced fairness and sustainability **in your experience today**? Why?

4. **Policy Insight:** If you were a fisheries manager for a coastal community, which strategy (or mix) would you recommend? Name **two concrete implementation steps** you'd propose (e.g., monitoring, enforcement, adaptive adjustment).

5. **Beyond Fisheries:** Give **one example** of another renewable common-pool resource (e.g., groundwater, forests, wildlife, carbon emissions). Briefly state which strategy might work best there and why.

Participation Check (instructor initials when reviewed)

☐ Recap table filled

☐ Five synthesis answers written

Turn this in to receive completion credit for today's activity.